

Agentic AI: Foundations and Open-Source Practice

Day 2 Task

1. Task Details

Task title	Reasoning and Acting: Comparing Direct Prompting, Chain-of-Thought, and ReAct on a Scenario of Your Own
Unit	Unit 1: Foundations of AI Agents — sub-topics 1.3 and 1.4 (Chain-of-Thought, the ReAct cycle)
Task type	Individual conceptual analysis, based on a scenario you choose yourself
Format required	A GitHub repository containing your project code, an output-screenshots folder, and a written analysis in Markdown
Submission	Link to your GitHub repository

2. Aim

Choose a small scenario of your own — one that has at least one question needing external information (a tool) and a few questions that need only careful reasoning — and demonstrate how the same kind of problem is handled by direct prompting, Chain-of-Thought prompting, and a ReAct agent. Explain how each approach works, what it can and cannot do on its own, and where its limitations show up. Compare the three approaches based on reasoning depth, use of tools, reliability, transparency of the answer, cost/speed, and consistency across repeated runs, and analyze which approach is most suitable for your chosen scenario and why. Your analysis should clearly demonstrate that direct prompting gives an answer straight away with no visible reasoning, Chain-of-Thought makes the model reason step by step before answering but still cannot fetch facts it doesn't have, and ReAct interleaves Thought, Action and Observation so the agent can call tools and use their results before reaching a final answer. Conclude by explaining when direct prompting, Chain-of-Thought, or ReAct would be the most appropriate choice for different types of problems.

3. Result

Your submission must produce the following, for the scenario you chose:

3.1 Explanation of each approach

For each of the three approaches, explain:

- What kind of question it can and cannot answer correctly.
- Whether it uses tools, and if so, how it decides when to call one.
- How it arrives at a final answer, from the user's question to the response.
- Where its limitations show up on your specific scenario.

Your explanation must clearly show:

- Direct prompting answers immediately from the model's own knowledge, with no visible reasoning and no tool access.
- Chain-of-Thought makes the model reason step by step before answering, which improves multi-step reasoning but cannot supply facts the model does not already know.
- ReAct cycles through Thought, Action and Observation — it reasons about what it needs, calls a tool to get it, observes the result, and repeats until it can give a final answer.

3.2 Comparison table

Fill in this table for your scenario, comparing the three approaches on each basis listed:

Basis for comparison	Direct prompting	Chain-of-Thought	ReAct agent
Reasoning depth			
Tool usage			
Reliability on multi-step questions			
Transparency (can you see how it got the answer?)			
Speed / cost			
Consistency across repeated runs			

3.3 Self-consistency observation

Using one of your Chain-of-Thought reasoning questions, run it multiple times at a non-zero temperature and record the answers. State the majority answer, whether it was correct, and what happens when temperature is set to 0.

3.4 Suitability analysis

State which of the three approaches is most suitable for your chosen scenario, and justify the choice using points from your comparison table and your self-consistency observation above.

3.5 Conclusion

Close your analysis by explaining, in general, when direct prompting, Chain-of-Thought, or a ReAct agent would be the most appropriate choice, for different types of problems — not only for your own scenario.

4. Submission Guidelines

- Push your complete project code (tool functions, your direct-prompting/CoT comparison script, your self-consistency script, and your ReAct agent) to a GitHub repository.
- Inside the repository, create a folder named screenshots and place output screenshots of all three approaches running on your scenario in it.
- Inside the repository, create a file named analysis.md containing your full written analysis from Section 3 — the explanation of each approach, the filled-in comparison table, the self-consistency observation, the suitability analysis, and the conclusion — written out in paragraphs, not just bullet points.
- Submit the link to your GitHub repository.

Your analysis.md is what will be read and graded first, so make sure it stands on its own — write it so that someone who has not seen your code can still follow your reasoning end to end.